# AI Medical Report Assistant

**Domain: Healthcare AI / Medical Image Analysis**

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**Index**



## 1. Introduction

Artificial Intelligence has become an essential technology in modern healthcare, enabling faster and more accurate disease detection. Medical image analysis using Deep Learning can assist healthcare professionals by identifying diseases from medical scans with high accuracy.

This project, **AI Medical Report Assistant**, is an end-to-end AI application that analyzes **Chest X-ray images** to detect **Pneumonia** or **Normal** conditions using a Deep Learning model built with TensorFlow. Based on the prediction, an LLM (Large Language Model) generates a professional AI-assisted medical report. A Streamlit web application provides an easy-to-use interface for uploading X-ray images, viewing predictions, and downloading reports.

## 2. Problem Statement

Diagnosing pneumonia from Chest X-ray images requires experienced radiologists and can be time-consuming, especially in hospitals with a high patient load.

Manual diagnosis may lead to delays, while many healthcare facilities lack sufficient medical specialists. An AI-assisted system can support healthcare professionals by providing faster preliminary analysis and automatically generating structured medical reports.

## 3. Objective

The primary objectives of this project are:

- Develop a Deep Learning model for Chest X-ray classification.
- Detect whether an uploaded X-ray image is **Normal** or **Pneumonia**.
- Display the model's prediction confidence.
- Integrate an LLM to generate AI-assisted medical reports.
- Build an interactive Streamlit application for image upload and prediction.
- Provide downloadable AI-generated medical reports.

## 4. Tools & Technologies

| **Tool** | **Purpose** |
|---|---|
| Python | Data preprocessing and model development |
| TensorFlow / Keras | Deep Learning model using Transfer Learning |
| Streamlit | Interactive web application |
| OpenRouter GPT-4.1 Mini | AI-generated medical reports |
| NumPy | Numerical computations |
| Pillow (PIL) | Image processing |
| Matplotlib | Visualization of training results |
| Scikit-learn | Model evaluation metrics |
| Kaggle Dataset | Chest X-ray dataset |

## 5. Methodology

### 1. Dataset Collection

- Downloaded the Chest X-ray Pneumonia Dataset from Kaggle.
- Organized images into Train, Validation, and Test folders.

### 2. Data Preprocessing

- Resized images to **224 × 224** pixels.
- Converted images to RGB format.
- Normalized pixel values.
- Prepared data generators for training and validation.

### 3. Deep Learning Model

- Used **Transfer Learning** with TensorFlow.
- Added custom classification layers.
- Final layer uses **Sigmoid activation** for binary classification.
- Compiled the model using Adam optimizer and Binary Crossentropy loss.

### 4. Model Training

- Trained the model using the training dataset.
- Monitored training and validation accuracy.
- Saved the trained model as:

  pneumonia_model.h5

### 5. Model Evaluation

Evaluated the model using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC Curve

- AUC Score
- Validation AUC
- Confusion Matrix

**6. LLM Integration**

- Connected the application with **OpenRouter GPT-4.1 Mini**.
- Generated professional medical reports based on prediction results.

**7. Streamlit Application**

Developed a user-friendly interface allowing users to:

- Upload Chest X-ray images
- Predict disease
- View confidence score
- Generate AI medical reports
- Download reports

## 6. Data Description

**Dataset Source**

**Chest X-ray Pneumonia Dataset**

https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

**Dataset Statistics**

- Training Images: **5,216**
- Validation Images: **16**
- Testing Images: **624**

**Classes**

- NORMAL
- PNEUMONIA

**Image Format**

JPEG (.jpeg)

**Image Size**

Resized to **224 × 224**

## 7. Application Design

**Application Workflow**

1. Upload Chest X-ray Image
2. Image Preprocessing
3. Deep Learning Prediction
4. Confidence Score Calculation
5. AI Medical Report Generation
6. Download Report

**Streamlit Interface**

**Home Page**

- Project Title
- Upload Button

**Prediction Section**

- Uploaded Image
- Disease Prediction
- Confidence Score

**AI Report**

- Findings
- Impression
- Recommendation
- Disclaimer

**Download Button**

- Download AI Medical Report

## 8. Key Features

- Chest X-ray Classification
- Pneumonia Detection
- AI-generated Medical Report

- Confidence Score
- Downloadable Report
- Modern Streamlit Interface
- Transfer Learning Model
- LLM Integration

## 9. Challenges Faced

- Selecting an appropriate Transfer Learning model.
- Image preprocessing and normalization.
- Integrating TensorFlow with Streamlit.
- Managing Python environment dependencies.
- Connecting OpenRouter API for report generation.
- Handling indentation and runtime errors during development.
- Designing a responsive and attractive user interface.

## 10. Real-Life Applications

This application can be used by:

- Hospitals
- Clinics
- Radiologists
- Medical Students
- Healthcare Researchers
- Telemedicine Platforms
- AI-assisted Diagnostic Systems

## 11. Conclusion

The **AI Medical Report Assistant** demonstrates how Deep Learning and Large Language Models can work together to improve medical image analysis.By combining TensorFlow, Streamlit, and OpenRouter GPT-4.1 Mini, the application provides fast disease prediction and automatically generates professional AI-assisted medical reports.This project highlights the practical use of AI in healthcare while serving as a decision-support tool for medical professionals.

## 12. Future Scope

### 1. Multi-Disease Detection

Expand the model to detect multiple chest diseases such as:

- Tuberculosis
- COVID-19
- Lung Cancer

### 2. Improved Deep Learning Models

Experiment with:

- EfficientNetV2
- DenseNet121
- Vision Transformers (ViT)

### 3. PDF Medical Reports

Generate professionally formatted PDF reports.

### 4. Doctor Review System

Allow doctors to edit and approve AI-generated reports.

### 5. Real-Time Hospital Integration

Connect with hospital PACS and Electronic Health Record (EHR) systems.

## 13. References


**Kaggle Dataset**

https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

**TensorFlow Documentation**

https://www.tensorflow.org/

**Streamlit Documentation**

https://streamlit.io/

**OpenRouter API**

https://openrouter.ai/

**OpenAI Python SDK**

https://github.com/openai/openai-python